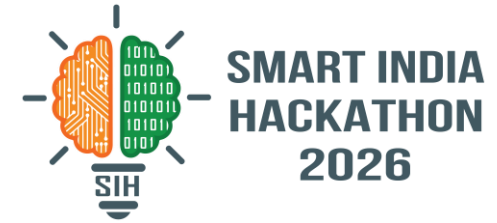


SMART INDIA HACKATHON 2026

Q-PRAMAAN



Explainable assurance for quantum digital signatures

PS ID **SIH26141**

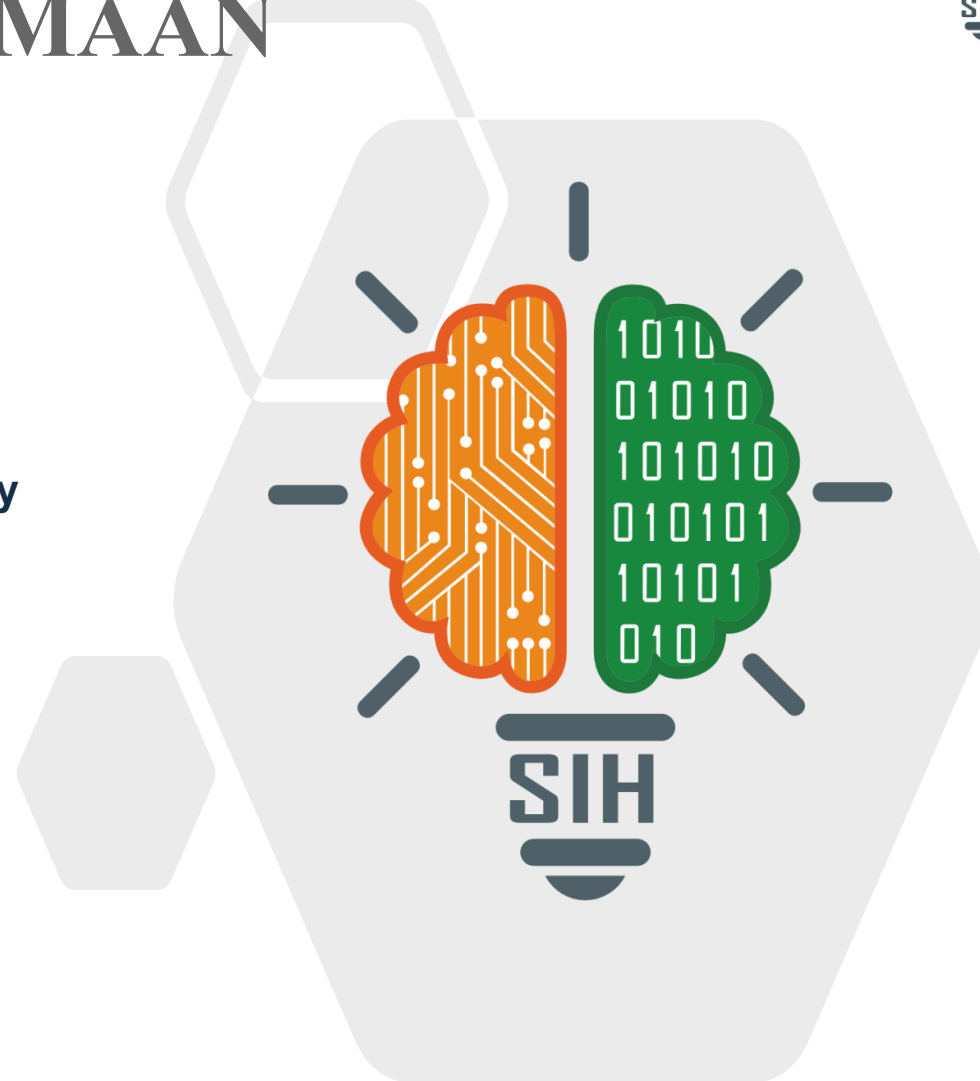
PROBLEM **Quantum-Inspired Cyber Threat
Detection for Digital Signature Security**

THEME **Blockchain & Cybersecurity**

CATEGORY **Software**

TEAM ID **125961**

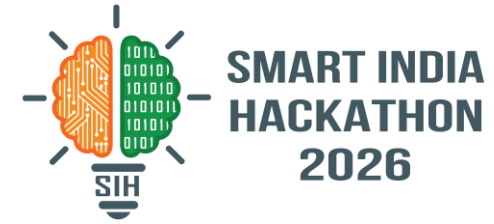
TEAM **Cache Hit**



Software simulation and security-assurance prototype

Cache
Hit

Q-PRAMAAN



Same 2% average error. One Bell outcome is manipulated.

Pooled detector · targeted channel

0 / 20,000

batches flagged

Ordinary channel: neither detector flagged a batch in 20,000 trials.

Conditional detector · targeted channel

19,967 / 20,000

batches flagged

PROPOSED SOLUTION

Simulate Alice–Bob–Charlie signatures over Bell-pair teleportation; verify the declaration and probe each teleportation outcome separately.

HOW IT ADDRESSES THE PROBLEM Authenticated sessions stop impersonation; an atomic ledger blocks replay; measurement thresholds expose channel anomalies.

INNOVATION A reproducible evidence trail separates signature validity, channel status and authorization.

An inspectable signature workflow

- 1 PREPARE** Alice creates BB84 signature states for approve / reject in a registered session.
- 2 TELEPORT** Simulated Bell pairs carry the states; recipients apply Pauli corrections and measure.
- 3 VERIFY** Bob accepts directly; Charlie checks a forwarded declaration with separate threshold.
- 4 DETECT** Six Pauli eigenstate probes, 20 fixed tests, loss checks and one-time access control.
- 5 EXPLAIN** Export counts, bounds, assumptions, decisions and a reproducible seed.

STACK Python · NumPy · SQLite · standard-library HTTP · JavaScript. No AI or model training.

Working prototype, measured against the same observations

20,000

batches per channel condition

Targeted channel: pooled 0 flags; conditioned detector 19,967 flags

Ideal protocol trials: 200 / 200 honest accepts; 0 / 200 forged accepts

FEASIBILITY

All 24 teleportation correction cases passed. The simulator, API, persistent replay guard, dashboard and evidence export run locally.

CHALLENGE AND RESPONSE

Physical noise, loss and calibration can mimic attacks. We report “insufficient evidence” when data are missing, quarantine model deviations, and display protocol assumptions beside every bound.

Finite-sample simulation; zero observed forgeries is not a security proof.

What a QDS engineer can decide from one report

SIGNATURE VALIDITY

The declaration passes or fails the direct / forwarded QDS rule.

CHANNEL STATUS

Conditional statistics show where the operating model breaks.

REQUEST AUTHORIZATION

Session binding, permissions and replay state control execution.

Resource planner selects a supported signature length and reports Bell pairs, classical bits and applicable bounds.

Users: QDS researchers, quantum-network developers and security evaluators.

Protocol basis and implementation references

Wallden et al. (2014)

BB84 QDS, symmetrization, forgery / repudiation analysis
arxiv.org/abs/1403.5551

Gottesman & Chuang (2001)

Foundational quantum digital signatures
arxiv.org/abs/quant-ph/0105032

Nadeem & Wang (2015)

Teleportation-based QDS research context
arxiv.org/abs/1507.03581

IBM Quantum Learning

Bell-pair teleportation and Pauli corrections
quantum.cloud.ibm.com/learning

Eagle-Eye QDS (PyPI)

Comparison baseline for explainable, statistical QDS tools
pypi.org/project/eagle-eye-qds/

Original contribution: condition-aware detection, assumption-linked bounds and repeatable adversarial evaluation.